

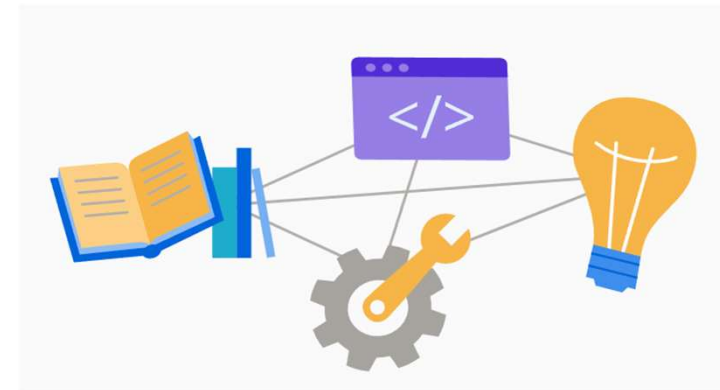
Module 1: .Net Overview

Overview

- .Net Platform Overview
- Common Language Runtime and JIT
- Assemblies, Modules and Resources
- .Net Tools (Compiler, Disassembler, IL Spy)

What is .Net

- Free, open-source, cross platform, high-performance application platform



- free = is for free
- open-source = developer in open
- cross platform = works & runs on Linux, MacOS and Windows
- application platform = programming languages & libraries

Supported languages and tools

- Languages

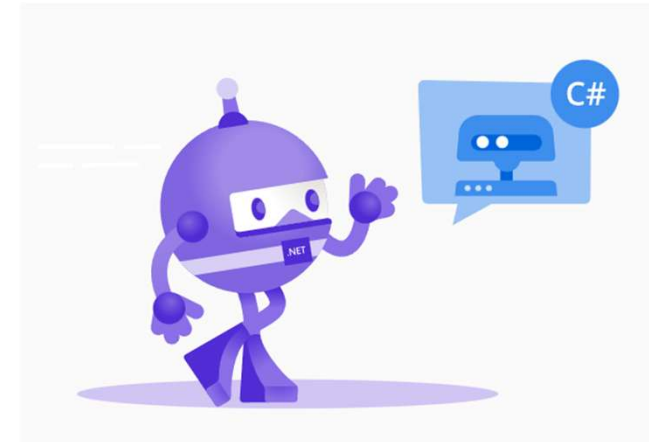
- C#
- F#
- VB.NET
- Others

- Tools

- Visual Studio
- Visual Studio Code

- Possibilities

- Web apps
- Mobile and Desktop apps
- Services
- Cloud, IoT, Microservices, etc



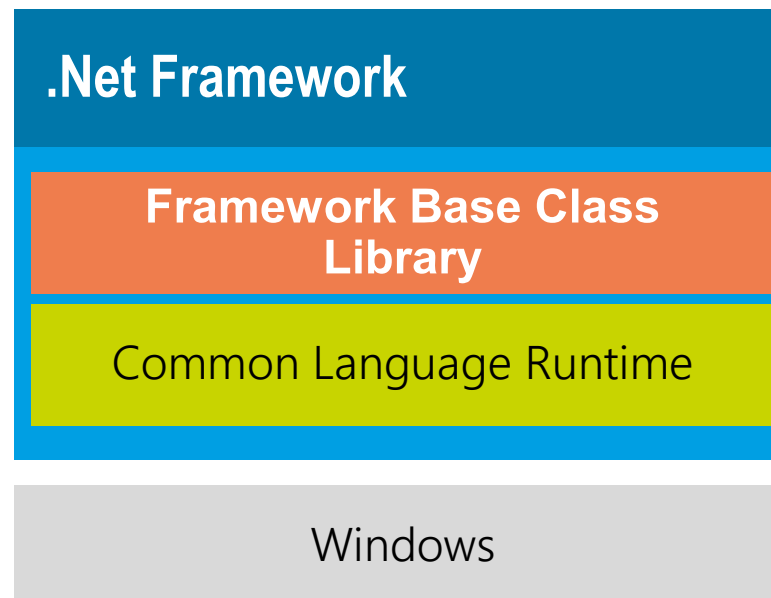
.Net Platform

- .Net Framework
- .Net Core / .NET 5 / 6 / 7 ...
- Mono for Xamarin



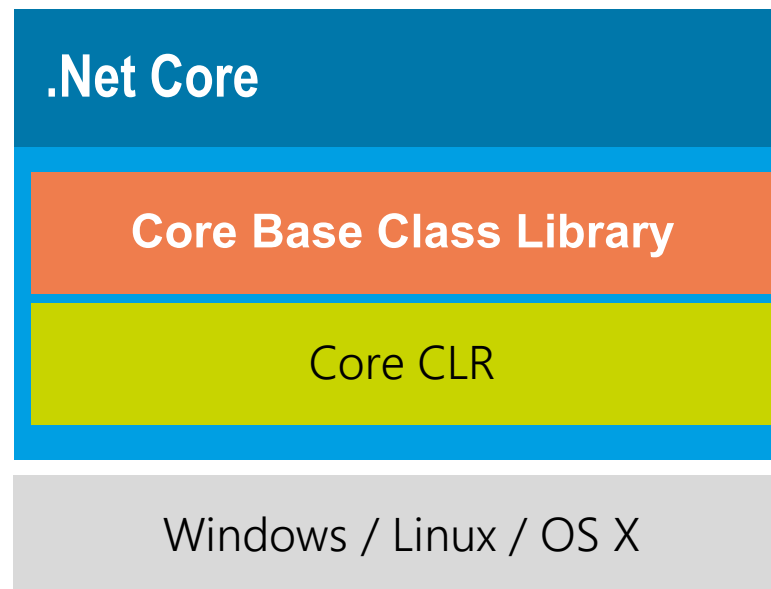
.Net Framework

- Released in 2002
- Windows centric - uses Windows API
- Must be preinstalled
- Partly open-source (can't contribute)
github.com/microsoft/referencesource



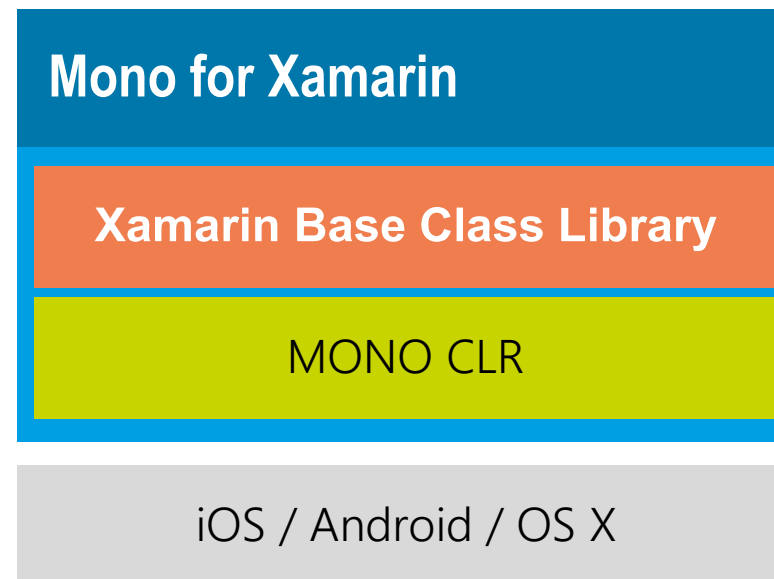
.Net Core

- Release in 2016 (lightwiegth and redesigned)
- Cross platform Linux, MacOS and Windows
- Does not replace .Net Framework
- Preinstalled / Self-Contained
- Completely open-source github.com/dotnet/core

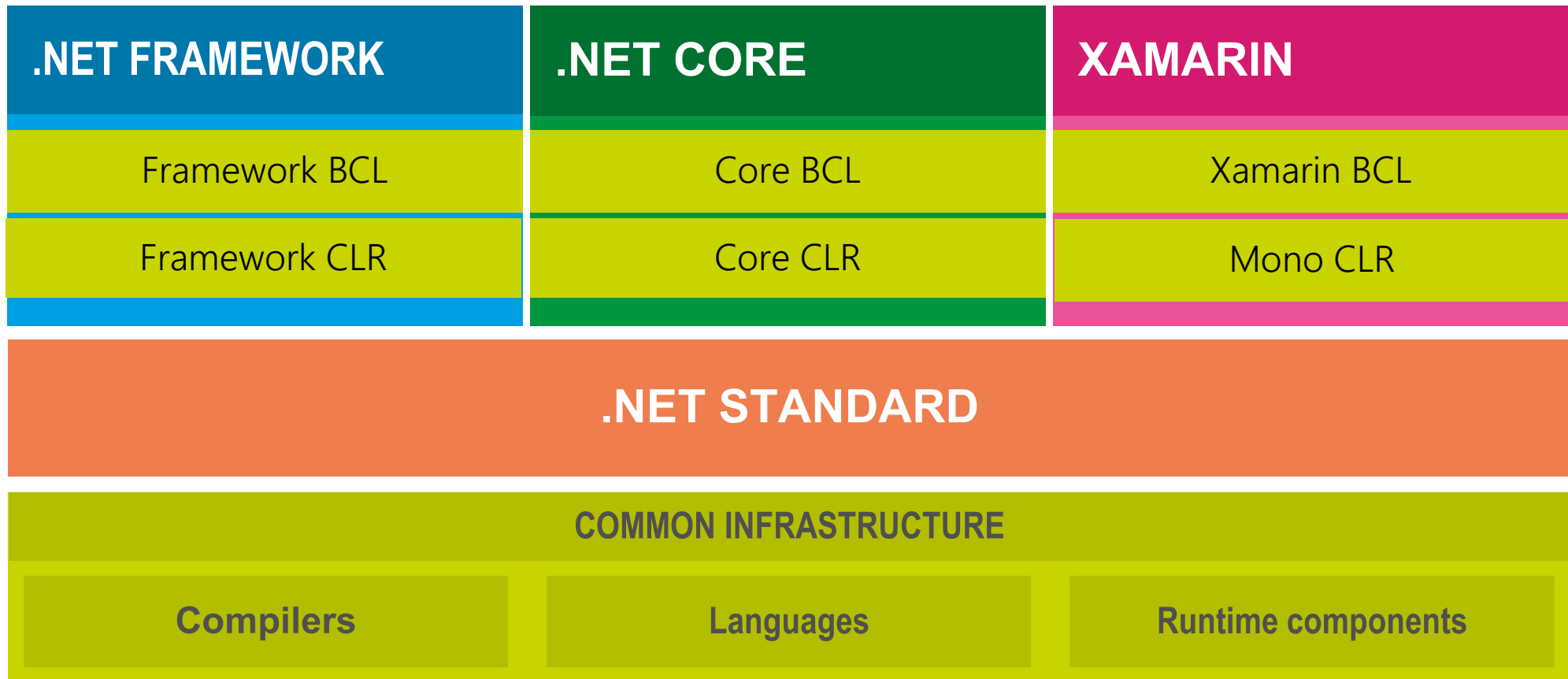


Mono for Xamarin / MAUI

- Release in 2011
- Mobile platforms Android, iOS, macOS, and Windows
- Native mobile and desktop apps with C# and XAML.
- Completely open-source
github.com/mono/mono



.Net Platform Overview

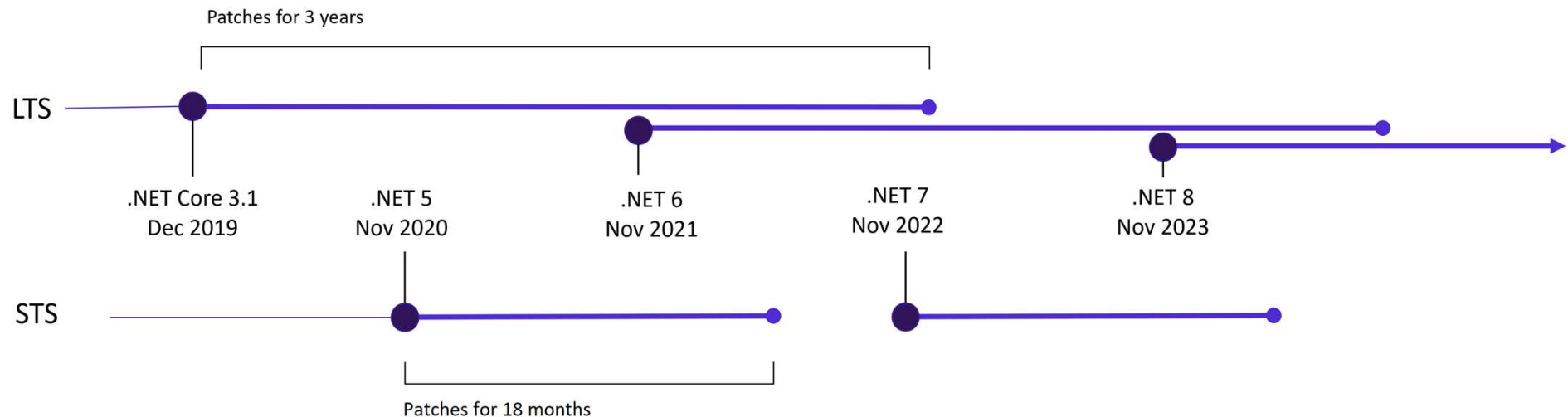


Note:

- *.Net 4.8 still integrated to W11*

.Net Support Policy

- **Standard Term Support** (STS) release that will be supported for 18 months
- **Long Term Support** (LTS) release that will be supported for 36 months



Note:

- *When support ends, Microsoft will stop releasing security updates, bug fixes, and other updates.*

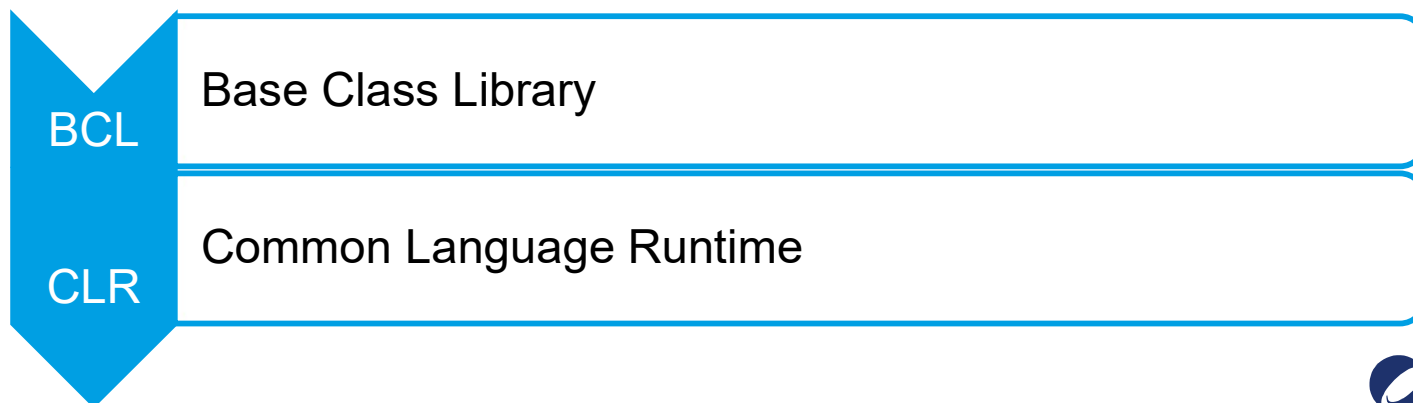
.NET consists of

- The Base Class Library (BCL)

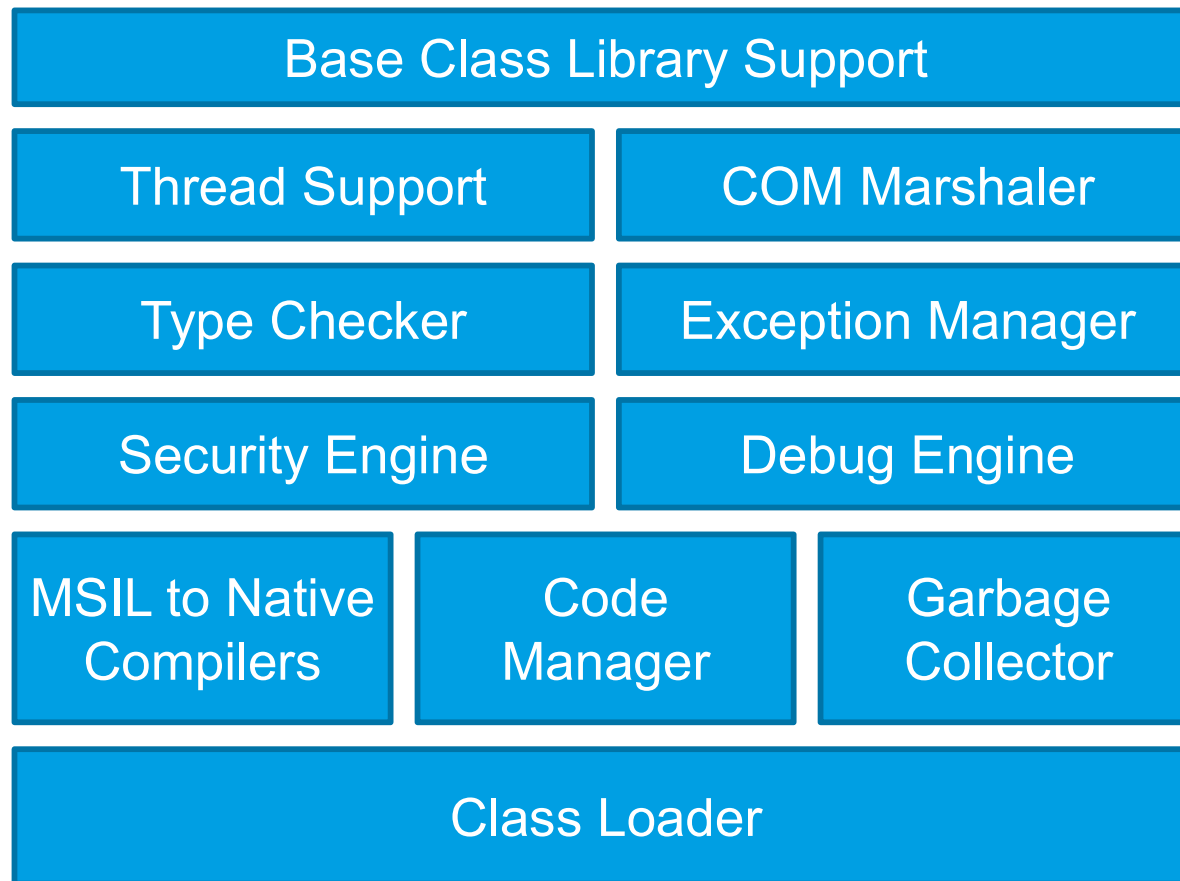
= set of classes, interfaces, and data types that provides basic functionality such as working with data types, I/O operations, buses (collections), data management and others

The Common Language Runtime (CLR)

= runtime environment in .NET that manages the execution of code, providing services like memory management, security, and exception handling, enabling multiple programming languages to work together on the same platform.



Common Language Runtime (CLS)



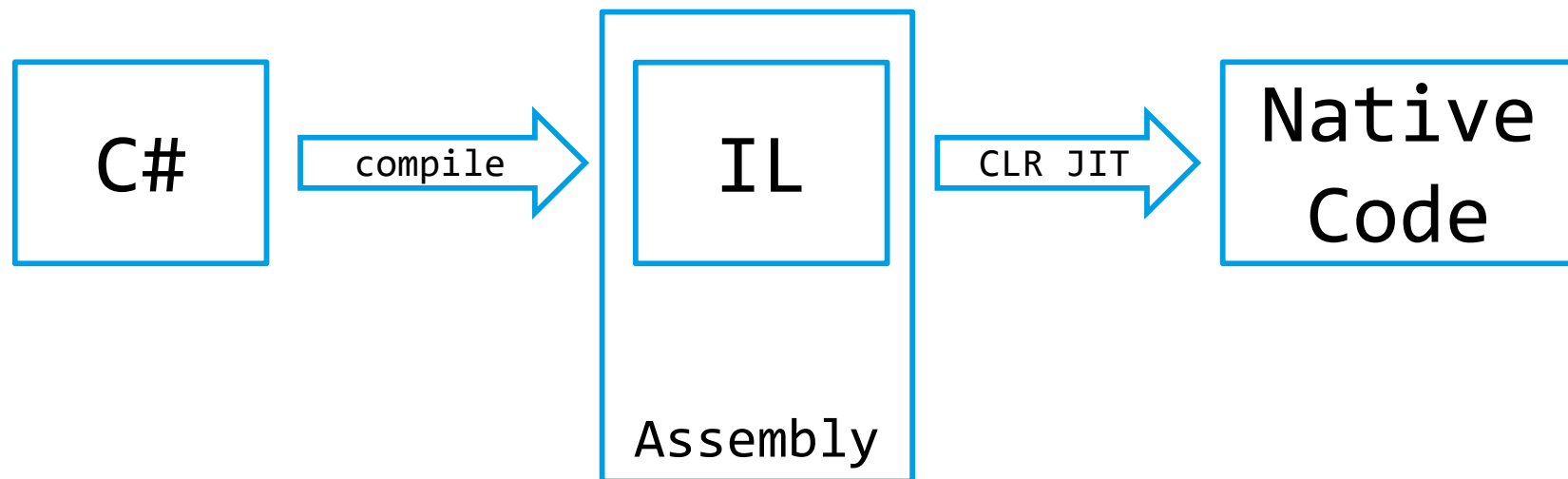
Code Execution in .Net

- **Intermediate Language (IL)**

= platform-independent, low-level code generated by .NET compilers from high-level language source code before execution

- **Just-In-Time (JIT) compilation**

= conversion of IL into native machine code at runtime, optimizing it for the specific environment



Assembly

- An assembly is a .dll or .exe file compiled using a .NET compiler
- Contains pseudo assembly code called IL
- Is the smallest deployment unit used, and is a component in .NET.

Assembly

PE Header

Manifest

Type Metadata

MSIL

Unmanaged Stub

Resources

.Net Framework SDK Tools

csc.exe

- C# compiler

ilasm.exe

- MSIL Assembler produces a final executable binary

ildasm.exe

- MSIL Disassembler inspects metadata and code of a managed binary

al.exe

- Assembly linker

Note:

- *Automatically installed tools with Visual Studio / .Net Framework SDK*

NET Core Command Line Tool

- Show help for dotnet tool

```
dotnet --help           // generic help
dotnet new --help       // detailed help
dotnet new list         // list of templates
dotnet new classlib --help // detailed help
```

- Create new project using dotnet tool

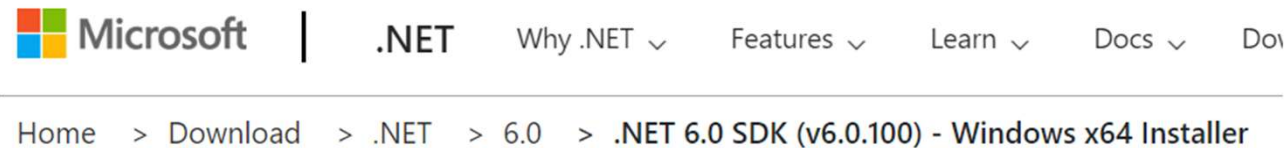
```
dotnet new classlib
    --framework "net6.0"
    --output HelloWorldCoreLib
```

- Build and run project

```
dotnet build
dotnet run
```

.Net Core SDK Tools

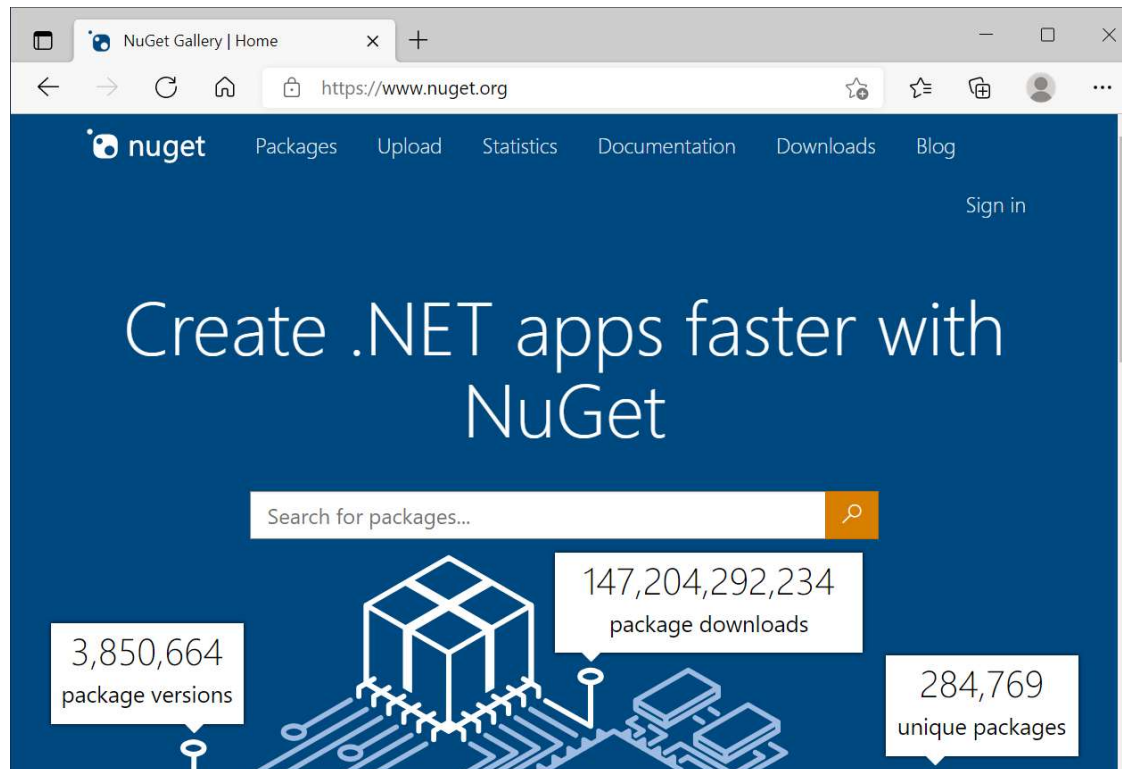
- The SDK includes the runtime and command line tools for creating .NET applications



Nuget

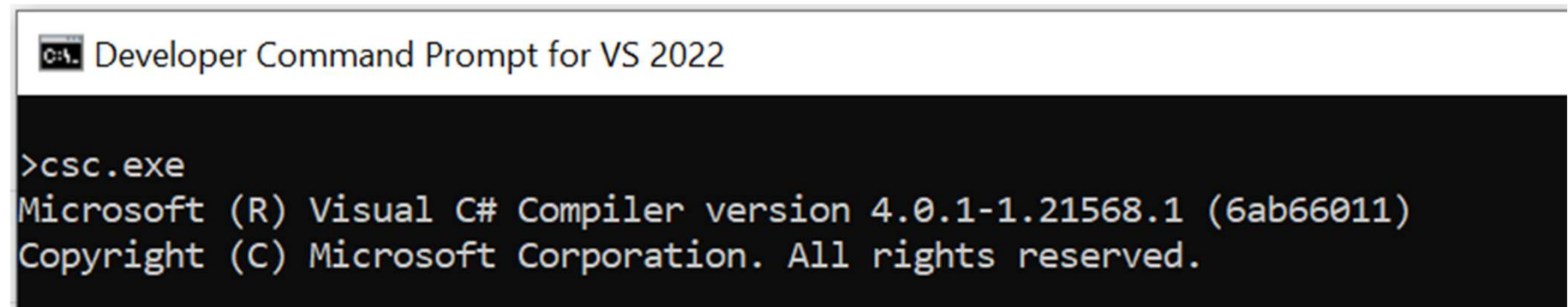
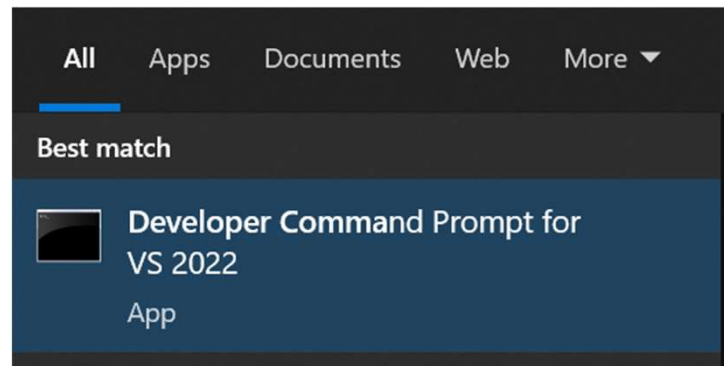
= package manager for .NET

- **nuget.exe** downloadable from www.nuget.org/downloads
- NuGet Gallery is the central package repository [nuget.org](https://www.nuget.org)



Visual Studio Tool: Developer Command Prompt

- Automatically sets the environment variables that enable you to easily use .NET tools

A screenshot of the 'Developer Command Prompt for VS 2022' terminal window. The title bar shows the icon and the text 'Developer Command Prompt for VS 2022'. The terminal content shows the command '>csc.exe' entered, followed by the output: 'Microsoft (R) Visual C# Compiler version 4.0.1-1.21568.1 (6ab66011)' and 'Copyright (C) Microsoft Corporation. All rights reserved.'

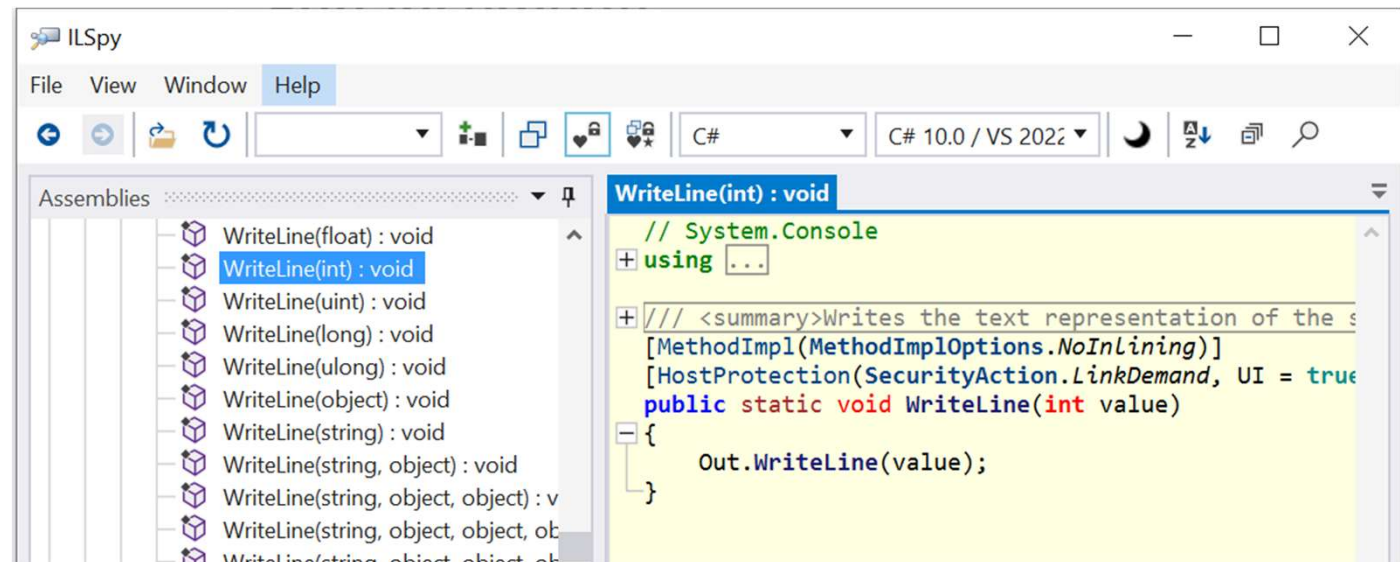
ILSpy

- Decompiles any .NET code
- Free to download from github.com/icsharpcode/ILSpy

ILSpy nuget v7.1.0.6543

ILSpy is the open-source .NET

Download: [latest release](#) | [late](#)



Other similar tools:

- *.Net Reflector / dotPeek / .NET CodeReflect*

Lab 01.1 – CmdLine and .Net SDK



Example:

1. Creating a ClassLibrary using the .Net SDK
2. Creating a Console App using the .Net SDK
3. Dissassembling using ILDASM
4. Dissassembling using ILSPY

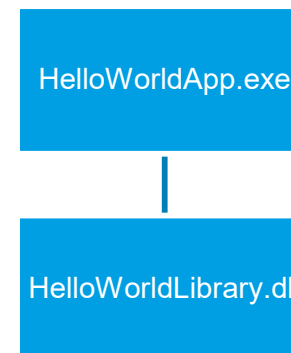
```
C:\Windows\System32\cmd.exe

Installing package 'Microsoft.CodeDom.Providers.DotNetCompilerPlatform' to 'C:\Users\Student\Documents\2125\HelloWorldDemo'.
GET https://api.nuget.org/v3/registration5-gz-semver2/microsoft.codedom.providers.dotnetcompilerplatform/index.json
OK https://api.nuget.org/v3/registration5-gz-semver2/microsoft.codedom.providers.dotnetcompilerplatform/index.json 534ms

Attempting to gather dependency information for package 'Microsoft.CodeDom.Providers.DotNetCompilerPlatform.3.6.0' with respect to project 'C:\Users\Student\Documents\2125\HelloWorldDemo', targeting 'Any,Version=v0.0'
Gathering dependency information took 25 ms
Attempting to resolve dependencies for package 'Microsoft.CodeDom.Providers.DotNetCompilerPlatform.3.6.0' with DependencyBehavior 'Lowest'
Resolving dependency information took 0 ms
Resolving actions to install package 'Microsoft.CodeDom.Providers.DotNetCompilerPlatform.3.6.0'
Resolved actions to install package 'Microsoft.CodeDom.Providers.DotNetCompilerPlatform.3.6.0'
Retrieving package 'Microsoft.CodeDom.Providers.DotNetCompilerPlatform 3.6.0' from 'nuget.org'.
Adding package 'Microsoft.CodeDom.Providers.DotNetCompilerPlatform.3.6.0' to folder 'C:\Users\Student\Documents\2125\HelloWorldDemo'
Added package 'Microsoft.CodeDom.Providers.DotNetCompilerPlatform.3.6.0' to folder 'C:\Users\Student\Documents\2125\HelloWorldDemo'
Successfully installed 'Microsoft.CodeDom.Providers.DotNetCompilerPlatform 3.6.0' to C:\Users\Student\Documents\2125\HelloWorldDemo
Executing nuget actions took 368 ms

C:\Users\Student\Documents\2125\HelloWorldDemo>csc.exe
'csc.exe' is not recognized as an internal or external command,
operable program or batch file.

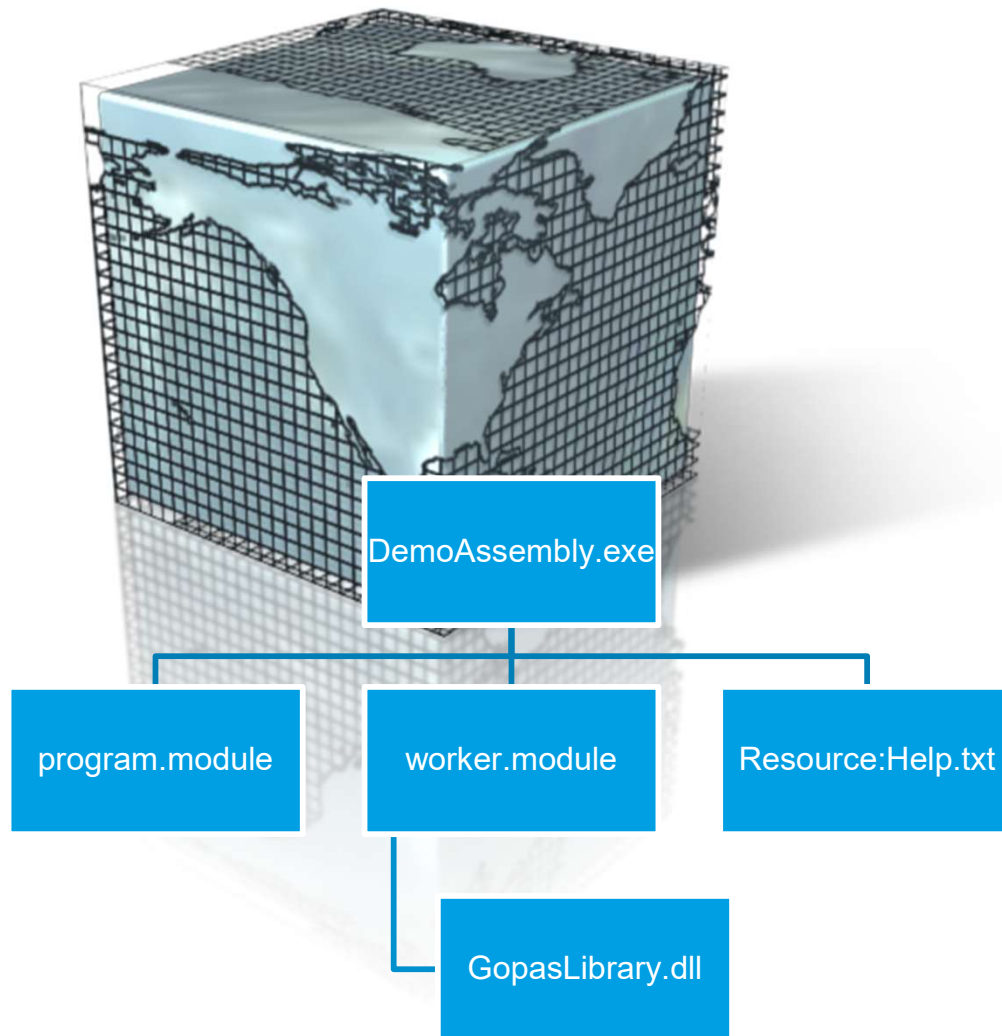
C:\Users\Student\Documents\2125\HelloWorldDemo>csc.exe
```



Review

- .Net Platform Overview
- Common Language Runtime and JIT
- Assemblies, Modules and Resources
- .Net Tools (Compiler, Disassembler, Assembly Linker)

Lab 01.1 - Compiling MultiFile Assemblies



Example:

1. Creating module **Library**
2. Creating module **Worker**
3. Creating module **Program**
4. Linking worker and program into the assembly
5. Updating module worker
6. Using Linked Resource